

Mid-semester Quiz

EGH445 Modern Control

September 7, 2020

Instructions: This quiz has a total of 3 problems, with several sub-problems.

The total number of possible marks for this quiz is 100. It is recommended that you attempt to answer all questions. All answers need to be justified. Explanations of the steps used are essential and will be carefully accounted for in the assessment. Best marks are awarded to numerically correct solutions, but partial credits may be given to partially complete solutions. Any assumptions made should be consistent with the question, clearly stated and justified.

Problem 1

Consider the system

$$\ddot{y}(t) + y(t) = -0.5\dot{y}(t) + u(t), \quad (1)$$

where $u(t)$ is the input and $y(t)$ is the (scalar) output.

A) Convert (1) into state-space form.

(10/100 marks)

B) Is the system (1) linear and time-invariant? (Justify your answer).

(10/100 marks)

C) Is the system (1) completely controllable? (Justify your answer).

(10/100 marks)

Problem 2

Consider again the system in Equation (1) of Problem 1 and the cost functional

$$J(x_0, u) = \int_0^\infty (6(y(t))^2 + 2(u(t))^2) dt$$

A) Find the optimal controller gain K in

$$u(t) = -K \begin{bmatrix} y(t) \\ \dot{y}(t) \end{bmatrix}$$

(25/100 marks)

B) Is the closed loop system obtained when using the optimal controller designed in point A) above stable?

(10/100 marks)

Problem 3

Consider a linear system described by:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ 10 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t)$$

A) Design a zero-state feedback regulator with desired closed-loop eigenvalues located at $\lambda_1 = -3$ and $\lambda_2 = -4$.

(20/100 marks)

B) Draw a block-diagram of the controlled system.

(15/100 marks)

END OF QUIZ